

Modbus RTU og Modbus TCP – Baggrundsforståelse

Formål

- Formål:

At give en grundlæggende forståelse af Modbus-protokollen (RTU og TCP).

Efter lektionen skal du kunne:

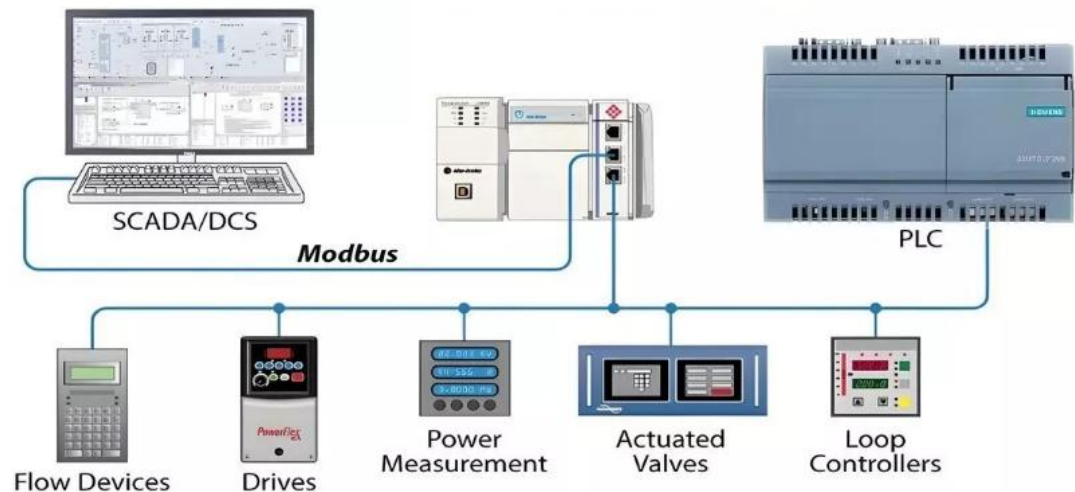
- Forklare forskellen mellem Modbus RTU og TCP.
- Forstå master/slave (client/server) princippet.
- Genkende Modbus-rammer og funktionkoder.
- Simulere kommunikation uden fysisk hardware.

Hvad er Modbus?

- Udviklet af Modicon (nu Schneider Electric) i 1979.
- Bruges til kommunikation mellem sensorer, PLC'er, SCADA, osv.
- Simpelt master/slave- eller client/server-system.

Eksempler:

- Sensorer, aktuatorer, PLC'er, SCADA-systemer.



Master/Slave-princippet

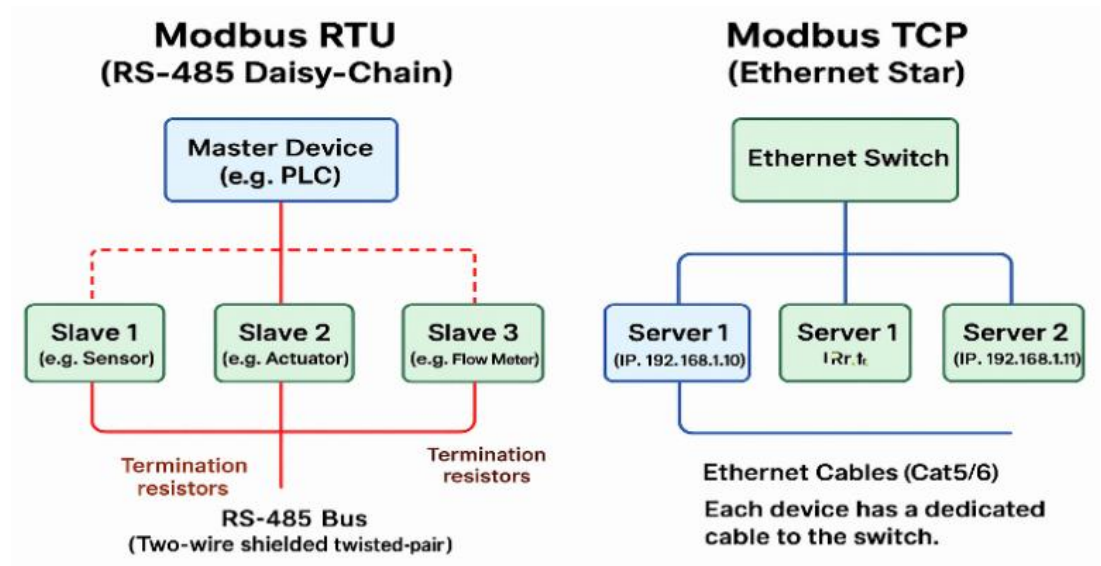
Struktur:

- Master (Client):
 - sender forespørgsler (fx ESP32, PLC).
- Slave (Server):
 - svarer (fx temperaturmåler).

Kun master initierer kommunikation.

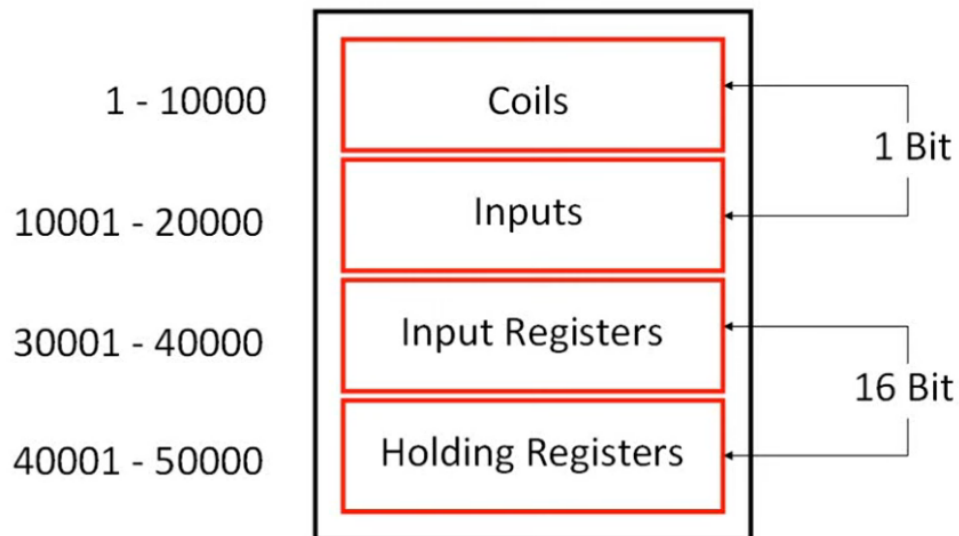
Eksempel:

- Master → “Send register 0”
- Slave → “Register 0 = 23.5°C”



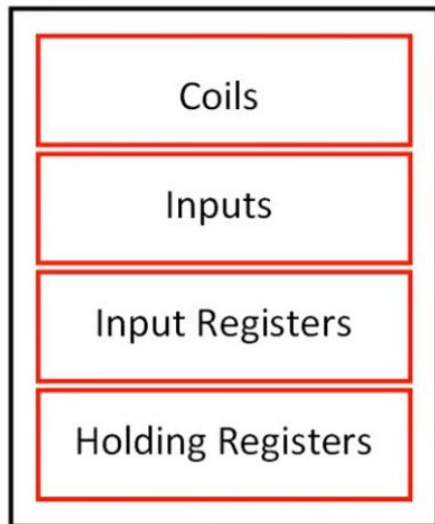
Datamodellen i Modbus

Modbus Memory Area - 4 Areas



Adressering og Funktionkoder

Modbus Memory Area - 4 Areas



Discrete Outputs

Discrete Inputs

Analog Inputs

Analog Outputs

Modbus Function Code		Purpose
Function Code 1	(Hex 0x01)	Read the status of output coils
Function Code 2	(Hex 0x02)	Read the status of input coils
Function Code 3	(Hex 0x03)	Read 16-bit values from holding registers
Function Code 4	(Hex 0x04)	Read 16-bit values from input registers
Function Code 5	(Hex 0x05)	Write to a single output coil
Function Code 6	(Hex 0x06)	Write a 16-bit value to a single holding register
Function Code 15	(Hex 0x0F)	Write to multiple output coils
Function Code 16	(Hex 0x10)	Write 16-bit values to multiple holding registers
Function Code 22	(Hex 0x16)	Modifies specific bits within a holding register

Modbus RTU – Kommunikation via RS485

RTU = Remote Terminal Unit

- Fysisk lag: RS485
- Maks. 32 enheder, op til 1200 m kabel
- Binær protokol, fejlkontrol via CRC

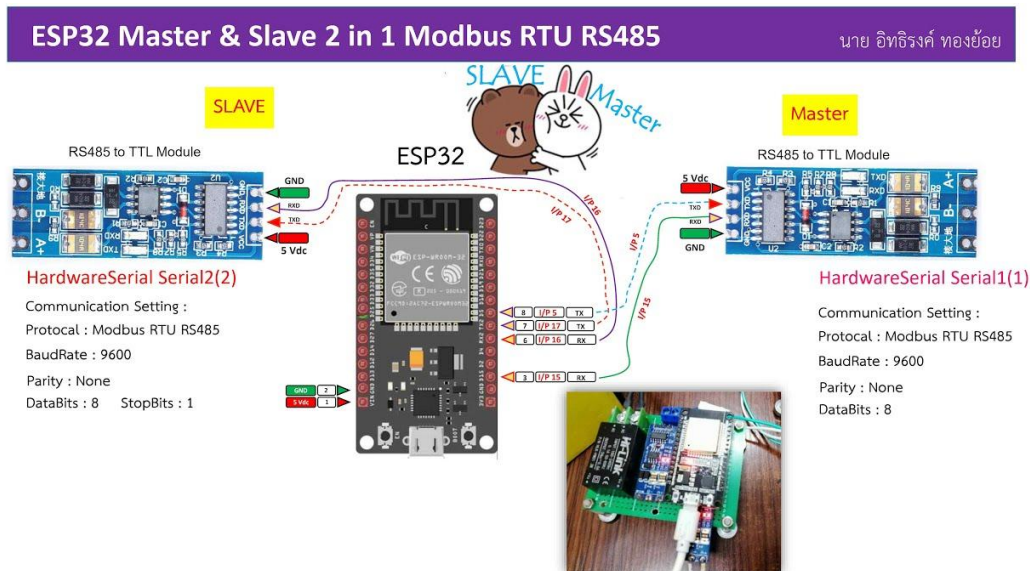
Eksempel (hex):

- 01 03 00 00 00 02 C4 0B

Fordel: stabil og enkel

Ulempe:

kræver fysisk kabel og retning-styring

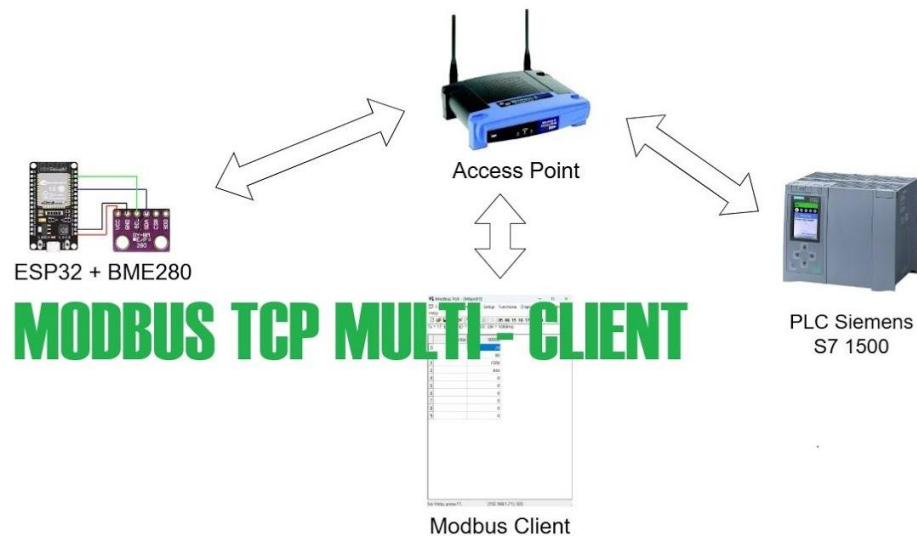


Modbus TCP – Kommunikation via Ethernet

- TCP = Transmission Control Protocol
- Kører over Ethernet/IP
- Samme struktur som RTU, men uden CRC
- Bruger port 502

Eksempel (hex):

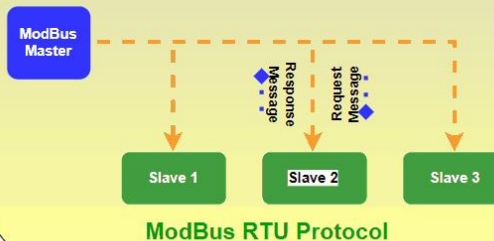
- 00 01 00 00 00 06 01 03
00 00 00 02
- Fordel: LAN/internet-kommunikation
- Ulempe: afhænger af netværk



Sammenligning – RTU vs TCP

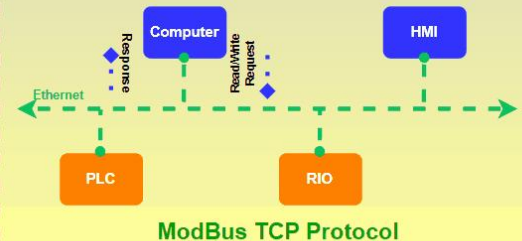
ModBus RTU

- **Protocol Type:** Serial communication (RS485 or RS232).
- **Network Topology:** Daisy chain (linked on two wires).
- **Maximum Range:** Up to 4,000 feet with RS485.
- **Speed:** Typically 38,400 baud rate.
- **Devices:** Pressure transmitters, flow meters, gas analyzers.
- **Use Case:** Field equipment, Data acquisition Devices

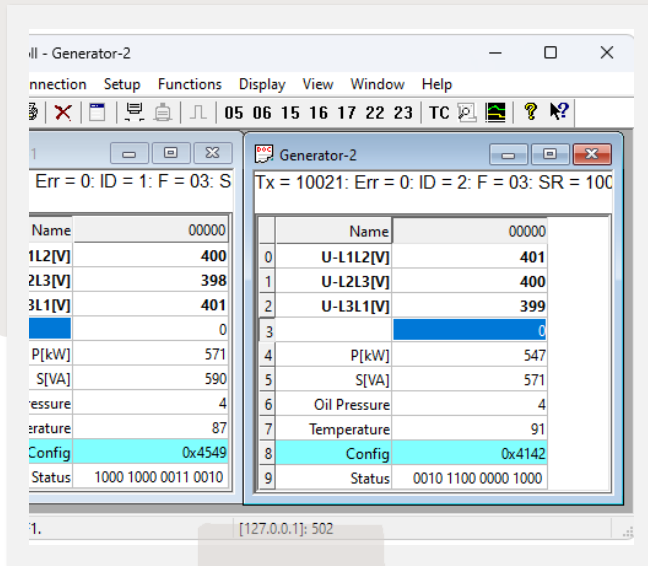


ModBus TCP

- **Protocol Type:** Ethernet communication. With RJ45 Connector
- **Network Topology:** Star, Ring, Tree, Line network.
- **Maximum Range:** Over existing Ethernet networks.
- **Speed:** Faster than Modbus RTU.
- **Devices:** PLCs, DCSs, Remote IOs, control room equipment.
- **Use Case:** High-level or remote applications.



Simulering – Modbus uden hardware



Softwareværktøjer:

- QModMaster – Master simulator
 - Gratis, open-source
 - RTU/TCP support

<https://sourceforge.net/projects/qmodmaster/>

- Modbus Slave – Slave simulator
 - Kommerciel demo
 - RTU/TCP simulering

https://www.modbustools.com/modbus_slave.html

- Node-RED + Modbus-noder
 - Visuel simulering over TCP
 - Ideel til IoT og dashboards

<https://nodered.org/>

• Download from Github Releases

- Visit the [Github Releases page](#) of ModbusTool.
- Choose ModbusTool.msi to download.
- Once the file is downloaded, double-click on it to initiate the installation process.
- Follow the on-screen instructions to complete the installation.

<https://github.com/ClassicDIY/ModbusTool>

<https://github.com/ClassicDIY/ModbusTool/releases>

Simuleringsøvelse

– Del A: Modbus RTU

1. Start QModMaster (Mode: RTU, Baudrate 9600, Slave ID 1)
2. Tilslut til Modbus Slave Simulator
3. Send “Read Holding Registers” (funktion 03)
4. Se svar i hex og noter CRC

Spørgsmål:

- Hvad sker, hvis du ændrer Slave ID?
- Hvordan ændres CRC?

Simuleringsøvelse – Del B:

Modbus TCP

- 1. Skift QModMaster til TCP Mode
- 2. IP: 127.0.0.1, Port: 502
- 3. Start Modbus Slave TCP Simulator
- 4. Send “Read Input Registers” (funktion 04)
- 5. Sammenlign frame-strukturen med RTU

Refleksion og forståelse

Drøft i grupper:

1. Hvornår bruges RTU fremfor TCP?
2. Hvad betyder CRC i RTU?
3. Hvordan påvirker byttet A/B kommunikation?
4. Hvordan kombineres RTU og TCP (gateway)?
5. Hvordan bruges Modbus-data i IoT?

Konklusion

Efter lektionen kan du:

- Forklare Modbus og dens anvendelse.
- Skelne mellem RTU og TCP.
- Genkende Modbus-rammer og funktionkoder.
- Bruge simulatorer uden hardware.
- Forstå kommunikation via RS485 og Ethernet.